operator **H** represents energy, we often call E_j the *energy eigenvalues* and $|E_j\rangle$ the *energy eigenvectors* of the system.

Exercise 4.5: Take any unit 3-vector $\vec{n}$ and form the operator

$$\mathbf{H} = \frac{\hbar\omega}{2}\sigma \cdot \vec{n}.$$

Find the energy eigenvalues and eigenvectors by solving the time-independent Schrödinger equation. Recall that Eq. 3.23 gives $\sigma \cdot \vec{n}$ in component form.

Let's suppose we have found all the energy eigenvalues E_j and the corresponding eigenvectors $|E_j\rangle$. We can now use that information to solve the time-dependent Schrödinger equation. The trick is to use the fact that the eigenvectors form an orthonormal basis and then expand the state-vector in that basis. Let the state-vector be called $|\Psi\rangle$ and write

$$|\Psi\rangle = \sum_j \alpha_j |E_j\rangle.$$

Since the state-vector $|\Psi\rangle$ changes with time and the basis vectors $|E_j\rangle$ do not, it follows that the coefficients α_j must also depend on time:

$$|\Psi(t)\rangle = \sum_j \alpha_j(t)|E_j\rangle. \qquad (4.29)$$

Now feed Eq. 4.29 into the time-dependent equation. The result is